

Power station

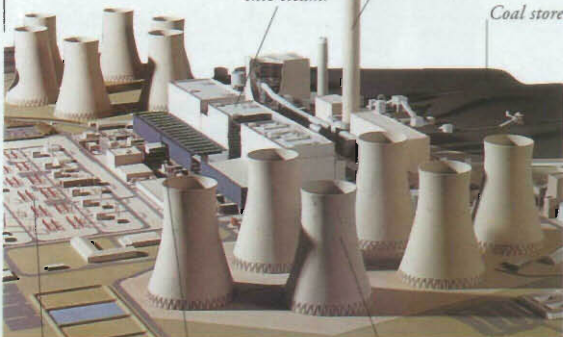
Most of the energy used in homes, offices, and factories is electricity produced by power stations. Inside a coal- or oil-fired power station, chemical energy stored within fuel turns into heat energy as the fuel burns in a furnace. The heat is used to boil water into steam, which drives turbines linked to electricity generators. The electricity reaches consumers via a network of cables called a grid.

Model of a coal-fired power station

Boiler house turns water into steam.

Fumes from furnace escape via chimney.

Coal store



Connections to electricity grid

The turbine house contains the turbines and generators.

Cooling towers turn steam from the boiler back into water.



The turbine blades revolve about 3,000 times each minute.

The generator uses the motion of the turbine to produce electricity.

Turbine-generator unit in a coal-fired power station

Turbine

A turbine is a machine powered by the force of moving liquid or gas. It consists of a set of angled blades mounted on a shaft. In a power station, jets of high-pressure steam strike the turbine blades and make them revolve at high speed. The turbine shaft is connected to an electricity generator. As the shaft spins, it turns an electromagnet inside the generator, producing an electric current.

Renewable energy

Energy that is produced without permanently using up the Earth's limited resources is called renewable energy. Apart from biomass fuels, which produce smoke and other fumes when burned, renewable energy sources are pollution-free, because they harness the energy of natural phenomena such as winds and waves. As the Earth's fossil fuel reserves are gradually used up, people will have to rely much more on renewable energy sources.



Geothermal power

Below the Earth's surface, water is turned into steam by geothermal energy – that is, the energy of hot, molten rocks. By drilling a well, this steam can be harnessed to drive generators. Electricity produced in this way is called geothermal power.



Hydroelectric power

A hydroelectric power station converts the kinetic energy of falling water into electricity. The power station sits under a dam at the end of a reservoir. Inside the power station, turbines and generators are driven by water rushing down with tremendous force from the reservoir above.



Wind power

A wind turbine is a tall tower with propeller-like blades that converts the kinetic energy of the wind into electricity. As the wind blows, the turbine's blades rotate and drive a small generator. A group of wind turbines is called a wind farm.



Wave power

Towers such as the one above stand in coastal waters and use the movement of the ocean's waves to produce electricity. As the waves rise and fall, they push a column of air inside the tower up and down. The to-and-fro motion of the air turns a turbine and drives a generator.



Solar power

Electricity produced from sunlight is called solar power. A "solar furnace" uses a vast bank of mirrors to focus sunlight on to water. The water boils into steam, which drives turbines and generators.



Biomass fuels

Plant material is called biomass. Millions of people around the world burn peat, wood, animal dung, and other biomass fuels to heat and light their homes, and to cook food. Burning biomass fuels releases chemical energy stored within the plant material.



Tidal barrage

At high and low tides, huge amounts of water move up and down river estuaries. A tidal barrage is a dam across an estuary. As the tides come in and go out, some water is allowed to pass through tunnels in the dam. The tidal flow drives electricity generators built into the dam.

Fossil fuels

Coal, oil, and natural gas are called fossil fuels, because they formed underground over millions of years from the fossilized remains of plants and animals. The Earth has limited supplies of these fuels, which cannot be replenished once exhausted.

Natural gas is made up of methane and small amounts of other gases.

Petrol, diesel, and many other fuels are made from oil.

Coal consists chiefly of the element carbon.



Natural gas

Oil

Coal

Charles Parsons

The engineer Charles Parsons (1854–1931) was born in London, England, of Irish parents. He is best known for inventing the steam turbine in 1884. Power stations around the world still use steam turbines based on Parson's designs. In 1897, his boat *Turbinia* became the first to use a steam turbine to power its propellers.



World energy use

Around 90 per cent of all the energy used comes from fossil fuels, which give out a lot of energy when burned, but release polluting gases into the air. Nuclear power is an alternative to fossil fuels, but produces dangerous radioactive waste. Hydroelectric power is the only form of renewable energy that is used in any significant amount.

